

How Love Flies: Emotional and Sonic Change in Popular Music, 1990–2022

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Abstract—

Index Terms—music information retrieval, emotion classification, BERT, Apache Spark, trend analysis, change-point detection

I. INTRODUCTION

A. Problem Statement

B. Contributions

II. RELATED WORK

A. Emotion and Sentiment in Song Lyrics

B. Audio-Feature and Valence Studies of Musical Trends

C. Trend and Change-Point Analysis of Cultural Time Series

III. DATASET

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B. Cleaning and Filtering

C. Per-Year Normalization and Corpus-Growth Bias

IV. METHODOLOGY

A. System Architecture and the Spark Pipeline

B. Audio-Feature Aggregation

C. Lyric Emotion Classification with BERT

D. Lexicon-Based Theme Detection

E. Unsupervised Era Discovery via K-Means

F. Statistical Analysis

1) Trend Testing:

2) Change-Point Detection:

3) Cross-Method Consensus of Change Years:

4) Verification of the Statistical Implementations:

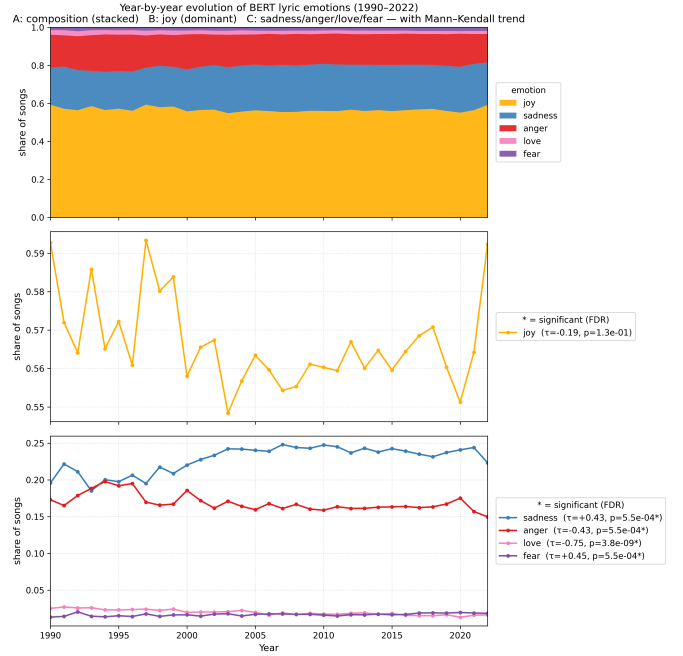


Fig. 1. Year-by-year evolution of the five BERT lyric emotions, 1990–2022.

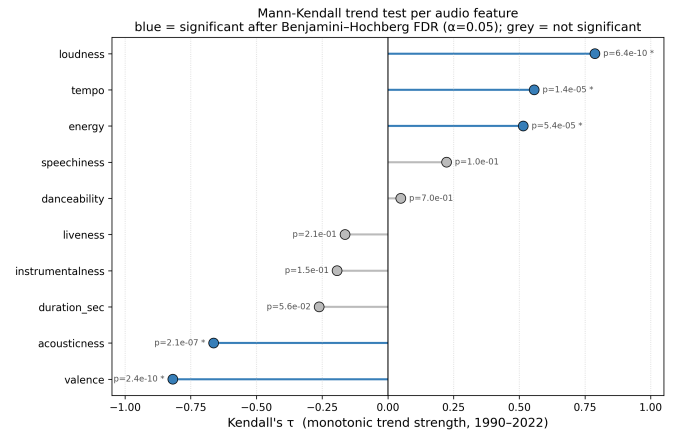


Fig. 2. Mann-Kendall trend strength per audio feature.

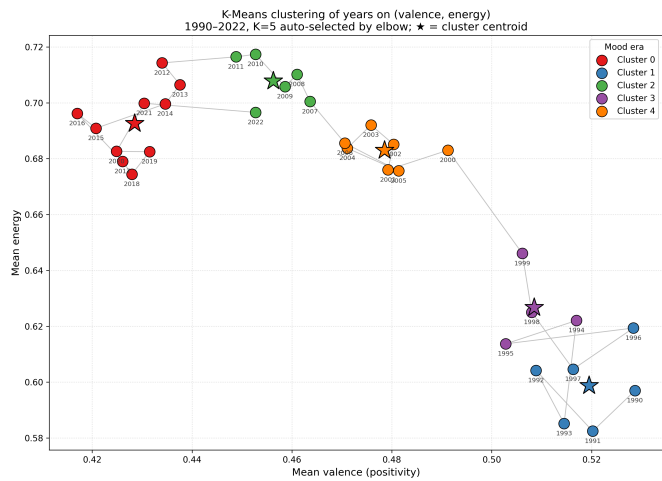


Fig. 3. K-Means clustering of years on (valence, energy).

TABLE I
TREND TEST RESULTS PER FEATURE

Feature	Kendall τ	p (FDR)	Direction

V. RESULTS

- A. Audio-Feature Trends
- B. Lyric-Emotion Trends Over Time
- C. Lyric Themes: Love, Lust, and Loneliness
- D. Discovered Eras: Mood and Genre Clusters
- E. Change Points and Cross-Method Consensus
- F. Performance and Scalability

VI. DISCUSSION

- A. Interpretation of the Observed Trends
- B. Convergent Validity Across Methods
- C. Threats to Validity and Limitations

VII. CONCLUSION AND FUTURE WORK

ACKNOWLEDGMENT

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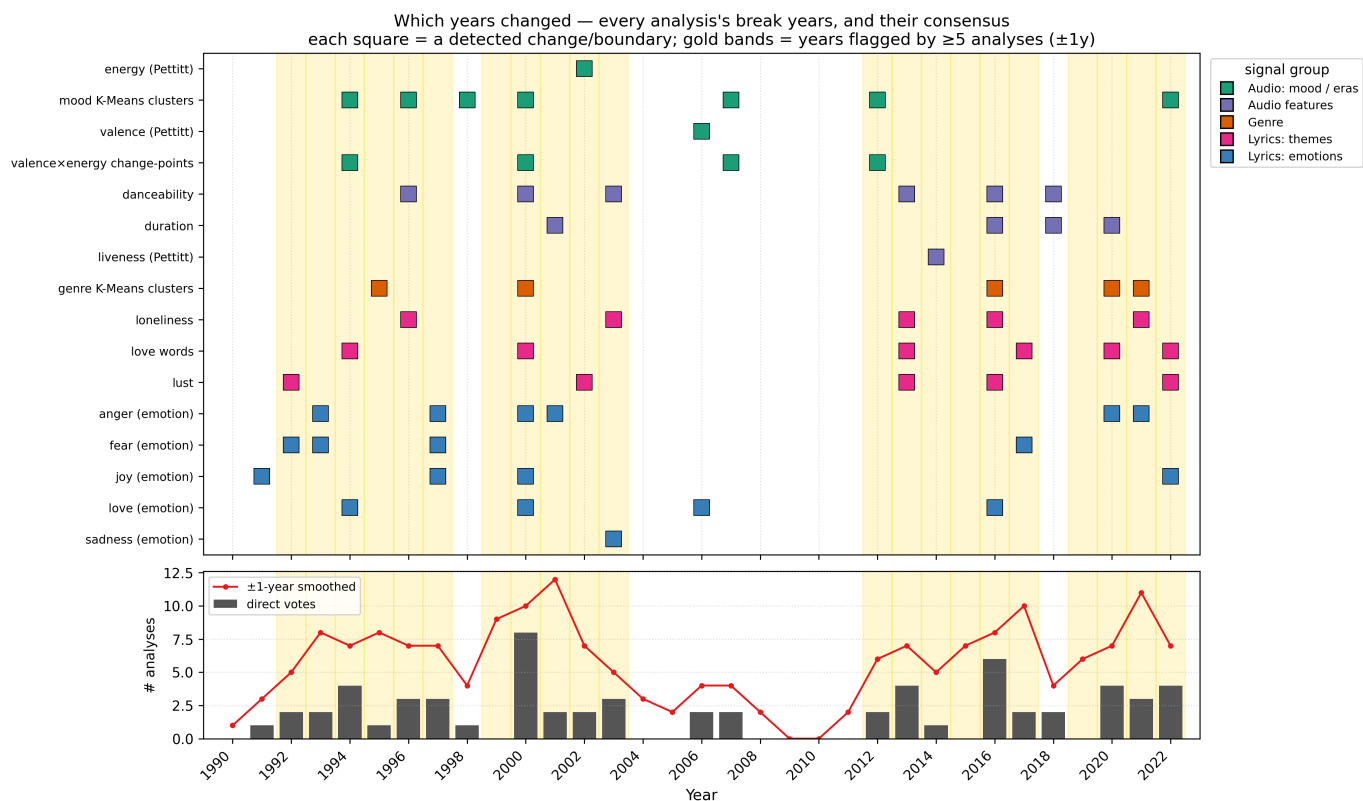


Fig. 4. Consensus of detected change years across every analysis.